Practical 40: Workflow Troubleshooting Challenge

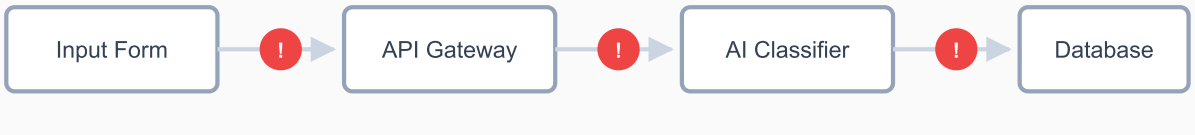
Identifying and resolving deliberate points of failure in an AI pipeline

In real-world automation, data pipelines frequently fail due to malformed inputs, network timeouts, or unexpected API responses. This challenge presents a supplied workflow containing six deliberate issues. Your task is to test the workflow, identify each failure, explain its root cause, and implement a suitable programmatic correction.

> **Instructions**
>
> Review the error matrix below. For each failure encountered during the testing phase, diagnose the symptom, determine the underlying cause within the script, and write the necessary validation or fallback logic to ensure the workflow can either recover gracefully or reject the payload with a descriptive error.

Failure Matrix Overview

Failure Point	Observed Symptom	Required Action
Missing Email	Database insertion fails with `NullConstraintException`.	Implement pre-flight payload validation.
Incorrect JSON	JSON decoder throws a parse error.	Add `try/except` block to catch bad formatting.
Invalid API Key	External AI service returns `401 Unauthorized`.	Implement credential verification prior to request.
Duplicate Form	Multiple identical records created in CRM.	Add a deduplication check using an idempotency key.
Failed Network	Script hangs or returns `ConnectionTimeout`.	Implement a retry mechanism with exponential backoff.
Unexpected Category	Routing engine crashes on unknown enum value.	Add a default fallback route for unknown labels.



Issue Diagnosis & Corrections (Part 1)

ISSUE 1: MISSING EMAIL ADDRESS

Symptom & Cause

The workflow attempts to insert a record into the CRM without verifying if the primary key (email) exists in the payload, resulting in a database constraint error.

CORRECTION

Implement early validation to reject the payload before processing.

```
def process_submission(payload):
    # Correction: Validate required fields immediately
    email = payload.get("email")
    if not email or "@" not in email:
        return {"status": "error", "message": "Valid email is required."}

    # ... proceed with processing ...
```

ISSUE 2: INCORRECT JSON FORMAT

Symptom & Cause

When the webhook receives malformed text instead of valid JSON, the standard `json.loads()` function throws a fatal `JSONDecodeError`, crashing the entire script.

CORRECTION

Wrap the parser in a `try/except` block to handle the error gracefully.

```
import json

def parse_webhook(request_body):
    try:
        data = json.loads(request_body)
        return data
    except json.JSONDecodeError:
        # Correction: Catch parse errors safely
        return {"status": "error", "message": "Malformed JSON payload."}
```

ISSUE 3: INVALID API KEY

Symptom & Cause

The AI classification stage attempts an HTTP request to the LLM provider with an expired or missing

Issue Diagnosis & Corrections (Part 2)

ISSUE 4: DUPLICATE FORM SUBMISSION

Symptom & Cause

Users double-clicking the submit button trigger the workflow twice. Because there is no check for existing records, duplicate entries are created in the database.

CORRECTION

Utilize an idempotency key (like a unique submission ID) to prevent duplicates.

```
def save_to_db(record, submission_id):
    # Correction: Check if submission_id already exists
    if database.exists({"sub_id": submission_id}):
        return {"status": "skipped", "message": "Duplicate submission detected."}

    database.insert(record)
    return {"status": "success"}
```

ISSUE 5: FAILED NETWORK REQUEST

Symptom & Cause

The external classification API occasionally drops connections. The script currently fails immediately on the first timeout, resulting in lost data.

CORRECTION

Implement a basic retry loop to handle transient network hiccups.

```
import time

def fetch_with_retry(url, retries=3):
    for attempt in range(retries):
        try:
            return requests.get(url, timeout=5)
        except requests.exceptions.Timeout:
            if attempt == retries - 1: # Last attempt
                raise Exception("Network request failed after retries.")
            time.sleep(2 ** attempt) # Exponential backoff
```

ISSUE 6: UNEXPECTED AI CATEGORY

Symptom & Cause